

Truncated Marginal Neural Ratio Estimation

A simulation-based inference approach

The problem: learning from a simulator

Bayesian inference asks which explanations remain plausible after data have been observed. If a model has parameters θ and the observation is $\mathbf{x}_0$, Bayes' theorem is

$$p(\theta \mid \mathbf{x}_0) = \frac{p(\mathbf{x}_0 \mid \theta)p(\theta)}{p(\mathbf{x}_0)}. \quad (1)$$

The prior describes plausible parameters before the observation. The likelihood describes how compatible the observation is with a proposed parameter value. The posterior is the updated distribution.

In many physical problems the forward model is available as a simulator. A parameter value can be passed through physical processes, measurement effects, and noise:

$$\theta \longrightarrow \text{simulator} \longrightarrow \mathbf{x}.$$

The difficulty is that the probability density of a particular output may not be available in a tractable form. Latent variables and complicated stochastic steps would have to be integrated out in order to evaluate the likelihood, drastically increasing the number of requisite calculations and quantity of required data.

- A simulator answers: "What data could this model produce?"
- A likelihood answers: "How probable is this dataset under the model?"
- TMNRE is designed for situations in which the first question is feasible but the second is not due to simulator complexity (like branching decisions, many hidden variables, etc.).

Speaker notes The distinction is between generating data and assigning an exact probability to data. A model may produce realistic mock observations while remaining unsuitable for ordinary likelihood-based MCMC.

Simulation-based inference

Simulation-based inference, or SBI, replaces explicit likelihood evaluations with simulated examples:

$$\theta^{(i)} \sim p(\theta), \quad \mathbf{x}^{(i)} \sim p(\mathbf{x} \mid \theta^{(i)}). \quad (2)$$

where $\theta^{(i)}$ is the i -th simulation.

Each pair records what the simulator produced for a known parameter choice. Traditional MCMC attempts to repeatedly generate parameter values that fit the dataset at random. SBI instead learns from a collection of simulated parameter-data pairs.

There is still a dimensionality problem. If the informative range is a fraction f of the prior range in each of d directions, the useful volume is approximately f^d . For $f = 0.1$ and $d = 20$,

$$(0.1)^{20} = 10^{-20}.$$

Random draws via MCMC from the full prior will therefore rarely reach the region that contributes most to the posterior.

Speaker notes The difficulty is not simply that there are many parameters. The useful region becomes a very small part of the space in which simulations are initially drawn.

The likelihood-to-evidence ratio

TMNRE estimates

$$r(\mathbf{x} \mid \theta) = \frac{p(\mathbf{x} \mid \theta)}{p(\mathbf{x})} = \frac{p(\theta \mid \mathbf{x})}{p(\theta)}. \quad (3)$$

A ratio greater than one favours a parameter value relative to the prior; a ratio less than one disfavors it.

The ratio can be learned as a classification problem. A joint pair is generated by drawing a parameter and simulating data from that same parameter:

$$(\theta, \mathbf{x}) \sim p(\theta)p(\mathbf{x} \mid \theta).$$

An independent pair combines independently drawn parameters and data:

$$(\boldsymbol{\theta}, \mathbf{x}) \sim p(\boldsymbol{\theta})p(\mathbf{x}).$$

The classifier predicts whether the two elements are associated. If $s(\mathbf{x}, \boldsymbol{\theta})$ is the probability assigned to the joint class, balanced classes give

$$\frac{s(\mathbf{x}, \boldsymbol{\theta})}{1 - s(\mathbf{x}, \boldsymbol{\theta})} = \frac{p(\boldsymbol{\theta}, \mathbf{x})}{p(\boldsymbol{\theta})p(\mathbf{x})} = \frac{p(\mathbf{x} | \boldsymbol{\theta})}{p(\mathbf{x})}. \quad (4)$$

The posterior can then be reconstructed, up to its normalisation, from

$$p(\boldsymbol{\theta} | \mathbf{x}_0) = r(\mathbf{x}_0 | \boldsymbol{\theta})p(\boldsymbol{\theta}).$$

Speaker notes The classifier is not being asked to name a physical object. It is being asked whether a parameter value and a dataset belong together. A difficult density-ratio problem is thereby turned into a supervised-learning problem.

Marginalisation: report the parameters of interest

A scientific model often contains parameters that affect the data but are not part of the final result. Write

$$\boldsymbol{\theta} = (\boldsymbol{\phi}, \boldsymbol{\psi}),$$

where $\boldsymbol{\phi}$ denotes parameters of interest and $\boldsymbol{\psi}$ denotes nuisance parameters. The required marginal posterior is

$$p(\boldsymbol{\phi} | \mathbf{x}_0) = \int p(\boldsymbol{\phi}, \boldsymbol{\psi} | \mathbf{x}_0) d\boldsymbol{\psi}. \quad (5)$$

Nuisance parameters remain present when simulations are generated. Their uncertainty is carried into the simulated data and therefore into the inferred distribution for $\boldsymbol{\phi}$.

TMNRE estimates selected one- and two-dimensional marginals rather than a single high-dimensional density.

- Parameters retained in a marginal are selected by the scientific question.
- Nuisance parameters remain in the simulator.
- The reported uncertainty includes their effect.

Speaker notes Marginalisation does not mean ignoring nuisance parameters. It means averaging over them. A cosmological parameter can therefore be reported while averaging over calibration, foreground, or source-model uncertainty.

Truncation: concentrate the simulations

TMNRE adds an iterative restriction of the prior so that later simulations are concentrated in regions that can explain the observation. The first round is broad and uses the original prior.

After this round, the marginal ratio estimators are evaluated using the observed dataset. They indicate which parameter values are supported by the observation and which are strongly disfavoured. The supported region is then retained for the next round.

If Γ is the retained region, the truncated prior is

$$p_{\Gamma}(\boldsymbol{\theta}) = \frac{p(\boldsymbol{\theta})\mathbb{I}_{\Gamma}(\boldsymbol{\theta})}{\int_{\Gamma} p(\boldsymbol{\theta}) d\boldsymbol{\theta}}, \quad (6)$$

where

$$\mathbb{I}_{\Gamma}(\boldsymbol{\theta}) = \begin{cases} 1, & \boldsymbol{\theta} \in \Gamma, \\ 0, & \boldsymbol{\theta} \notin \Gamma. \end{cases}$$

Where the denominator is a scaling factor, like in Bayes'.

The indicator function acts like a gate. It leaves values inside Γ available and blocks values outside it. The denominator renormalises the remaining prior so that it again integrates to one.

The truncation changes the range of the prior, but not its relative shape inside the retained region. For example, suppose a parameter originally has a uniform prior from 0 to 100, and the first round indicates that values from 35 to 65 contain essentially all of the relevant posterior mass. The next round samples only from 35 to 65. Values within that interval remain equally likely relative to one another; values outside it are no longer simulated.

The same applies to a non-uniform prior. If the original prior favours some values over others, truncation preserves those relative preferences inside the retained region. It does not replace the original prior with a new flat prior.

In one dimension, the retained region may be described by lower and upper bounds around the bulk of a marginal posterior. In several dimensions, the bounds are obtained from selected one- and two-dimensional marginal estimates. A conservative threshold removes low-density tails without discarding a separate mode or a region that still contains meaningful probability.

The next round repeats the sequence: draw parameters from the truncated prior, run the simulator, train the classifiers again, and update the retained region. The process stops when further rounds barely change the region.

Truncation is a compromise between coverage and efficiency. Keeping the entire prior protects against missing a possible explanation but wastes simulations. Truncating too aggressively saves simulations but can permanently remove a genuine explanation. TMNRE uses the initial broad search and conservative updates to balance these risks.

Speaker notes A useful analogy is searching for a small object in a large room. The first search covers the whole room. Once the likely area has been identified, later searches spend more time there. The search area can be reduced, but it must remain large enough to include every location that is still reasonably plausible.

A region is useful because it contains meaningful posterior probability, not simply because it contains the highest point of the curve. A broad region with moderate density may contain more total probability than a narrow peak. Truncation should therefore use estimated posterior mass and checks for alternative modes, rather than an arbitrary cutoff chosen only for convenience.

The TMNRE sequence

1. Specify the simulator, prior, observed data, and parameters of interest.
2. Draw an initial simulation set from the original prior.
3. Train neural classifiers for the selected marginal parameter sets.
4. Evaluate the ratio estimators on the observation.
5. Construct a conservative truncated region.
6. Draw a new simulation set from the truncated prior.
7. Retrain and repeat until the region changes very little.
8. Produce the requested marginal posteriors.
9. Test calibration and coverage using additional simulations.

Marginalisation reduces the complexity of the inferential target. Truncation reduces the number of simulations spent in regions already shown to be implausible.

Application: the cosmic microwave background

The CMB is used to constrain cosmological parameters, but analyses can also contain many nuisance parameters. Cole et al. apply TMNRE to CMB inference and compare it with conventional MCMC. Their results show that converged one- and two-dimensional posteriors can be obtained with orders of magnitude fewer simulator calls in the studied problem. The required number of simulations is effectively independent of the number of nuisance parameters because TMNRE targets marginals rather than the full joint posterior.

The application also demonstrates local amortisation. A trained estimator can be evaluated for many nearby mock observations. This makes coverage and consistency tests practical: a nominal 68 percent credible region should contain the true parameter approximately 68 percent of the time over repeated simulated experiments.

Application: strong gravitational lensing

A strong-lensing image contains a background source distorted by a foreground galaxy. Dark-matter subhalos in the lens and along the line of sight produce smaller perturbations. These perturbations may be difficult to identify individually, while their population can carry information about dark matter.

Anau Montel et al. combine parametric lensing models with TMNRE to constrain the cutoff scale of the warm-dark-matter halo mass function from multiple simulated Hubble-resolution images. The method marginalises over realistic perturbers and over uncertainty in lens and source parameters.

Coogan et al. examine individual subhalo measurements while including populations of hundreds of subhalos and line-of-sight halos. TMNRE returns marginal posteriors for selected subhalo parameters without requiring every perturber to appear in the final plot.

- The data are images rather than simple numerical summaries.
- The nuisance population is physically part of the lensing model.
- Marginalisation retains its uncertainty without displaying every nuisance parameter.

Speaker notes Strong lensing shows why realistic nuisance modelling matters. Ignoring small perturbers could produce narrow constraints, but those constraints would not describe the physical system being observed.

Application: the sky-averaged 21-cm signal

The redshifted 21-cm signal from the Cosmic Dawn and Epoch of Reionization contains information about the first galaxies and the early universe. Its measurement is complicated by foreground emission, beam uncertainties, ionospheric distortions, and receiver noise.

Saxena et al. apply TMNRE to mock observations for the REACH experiment. Their simulations include beam-weighted foregrounds, a physically motivated 21-cm signal, and radiometric noise. Combining multiple time slices and antennas improves the constraints and exploration of the cosmological signal.

The target parameters can describe the astrophysical 21-cm signal, while foreground and instrument parameters are treated as nuisance quantities. They remain in the simulations. Once the first round identifies signal and instrument combinations compatible with the observation, later simulations can focus on that region.

Reliability and limitations

TMNRE addresses computational inefficiency but does not remove the need for model validation. The simulator determines which uncertainties can appear in the training set, and the prior determines the initial parameter space. If either is unrealistic, a well-trained classifier can still produce a misleading posterior.

The main risk of truncation is premature removal of a genuine posterior region. Conservative thresholds and checks for missed modes are therefore essential. Neural classifiers may also be imperfect in narrow or multimodal problems.

The locally amortised estimator enables repeated tests that are expensive for a new MCMC analysis. Useful checks include coverage tests, simulation-based calibration, posterior-predictive checks, and comparison with a trusted likelihood-based method.

Conclusion

TMNRE is a simulation-based inference method for models in which data can be generated more easily than their likelihood can be evaluated. Its three central ideas are neural ratio estimation, marginalisation over nuisance parameters, and iterative truncation of the prior.

The CMB application demonstrates efficiency and local amortisation. Strong-lensing applications demonstrate marginalisation over complex populations of dark-matter perturbers. The 21-cm application demonstrates the inclusion of foreground and instrument effects in a physically motivated simulator.

TMNRE is therefore not a replacement for careful modelling. It is a way to combine realistic simulators with efficient and testable inference when conventional likelihood-based methods become impractical.